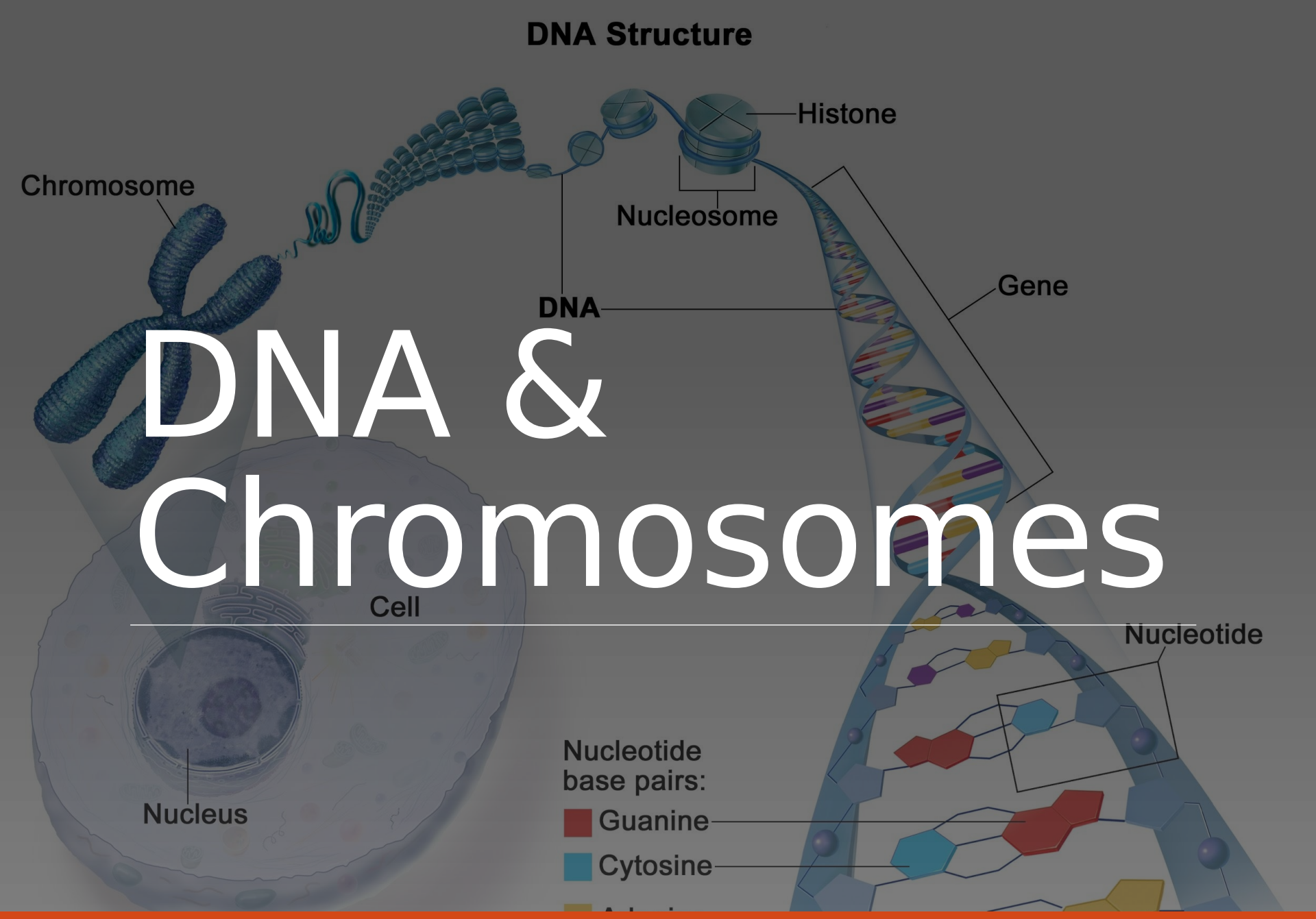


DNA & Chromosomes

OUTLINE

- Identification of DNA as the genetic (hereditary) material
- Structure of DNA and its implications
- Packaging of DNA in prokaryotes and eukaryotes
- General structure of chromosomes

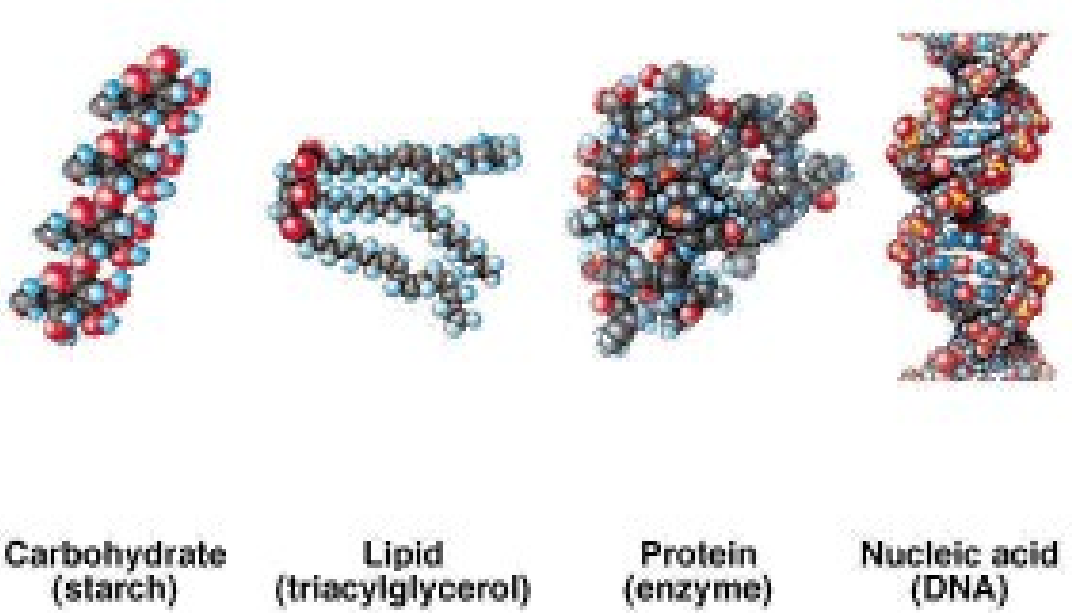
The Structural Basis of Cellular Information: DNA, Chromosomes, and the Nucleus

- Cells possess a set of “instructions” that specify their structure, dictate their functions, and regulate their activities (**Instructive**)
- These instructions can be passed on faithfully to daughter cells (**Hereditary**)
- These instructions change through generations that allow speciation and evolution. (**Dynamic**)

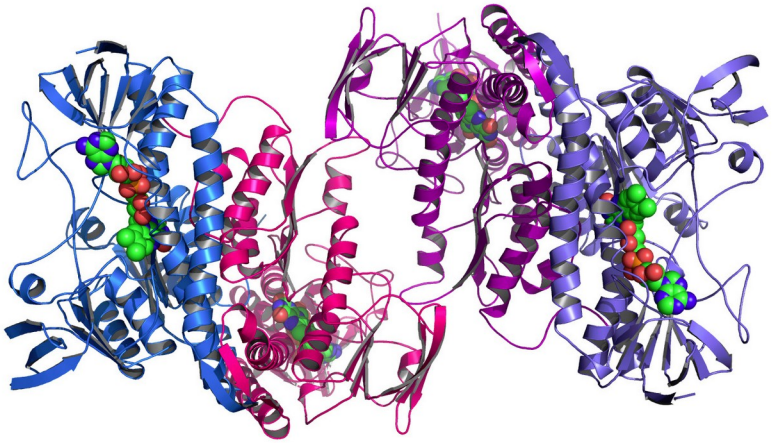
What is the nature of the genetic material?

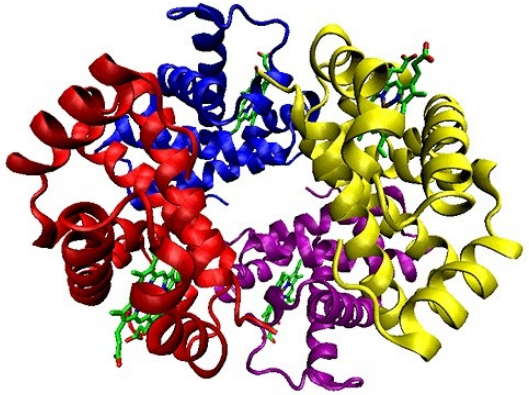
Think about the biological macromolecules;

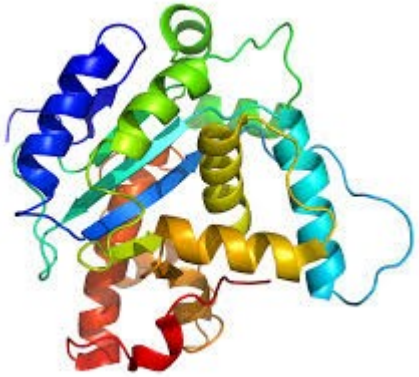
- Carbohydrates
- Lipids
- Proteins
- Nucleic acids

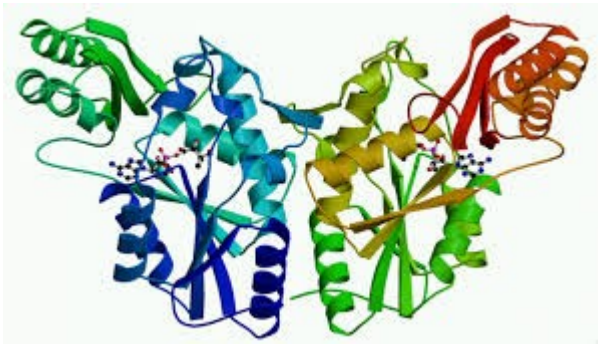


Proteins have variable structure and function









16.1 Chemical Nature of the Genetic Material

- In 1869, Johann Friedrich Miescher reported the discovery of the substance now known as DNA
  - He was interested in the chemistry of the nucleus, which most the scientists guessed was the site of the cell's genetic material.
- A few years later, Walther Flemming first observed chromosomes while st[illegible] the microscope

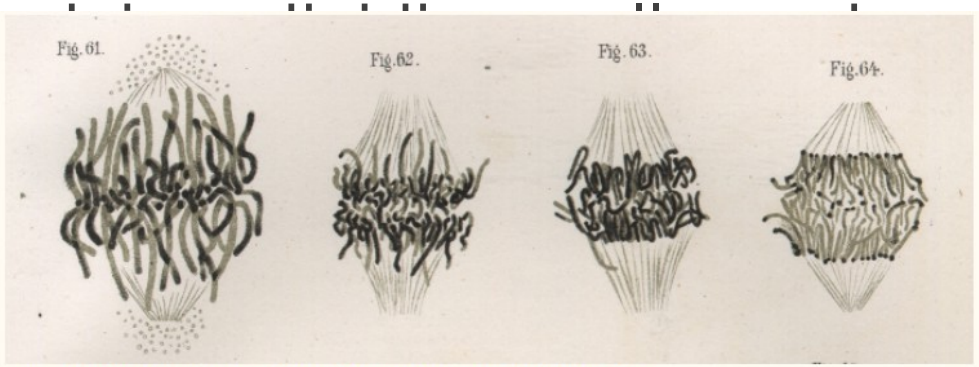


Figure 2 | **Illustration from *Zellsubstanz, Kern und Zelltheilung*[22].**

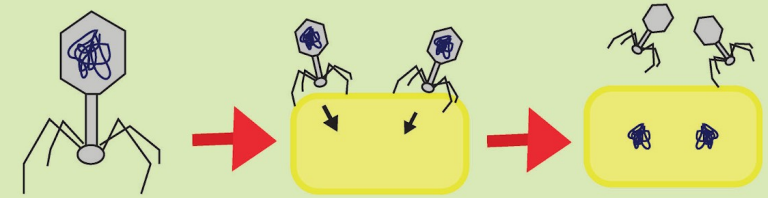
Flemming, W. Zellsubstanz, Kern und Zelltheilung (F. C. W. Vogel, Leipzig, 1882)

Genes and Protein

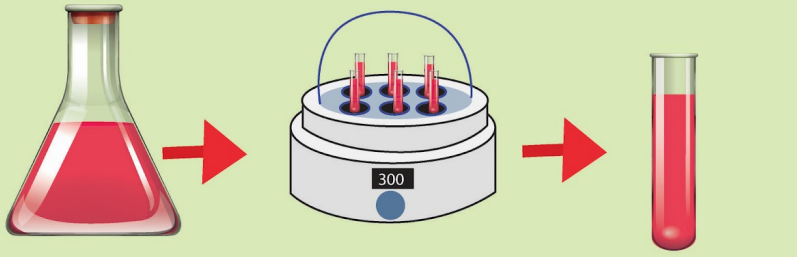
- From 1910 to the 1940s, most scientists believed that genes were made of protein rather than DNA
- This idea prevailed even after **two** important lines of evidence confirmed that DNA is the genetic material
  - Hershey and Chase's experiment: when bacteriophages, which are composed of DNA and protein, infect bacteria, their DNA enters the host bacterial cell, but most of their protein does not. Hershey and Chase and subsequent discoveries all served to prove that DNA is the hereditary material.
  - https://www.youtube.com/watch?v=Zh3pff--dgM

Hershey and Chase Experiment

Experiment 1: Testing Proteins



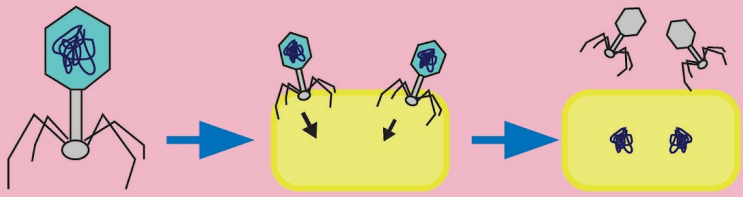
Protein coats radiolabelled → Bacteria infected → **NO** radioactivity enters cells



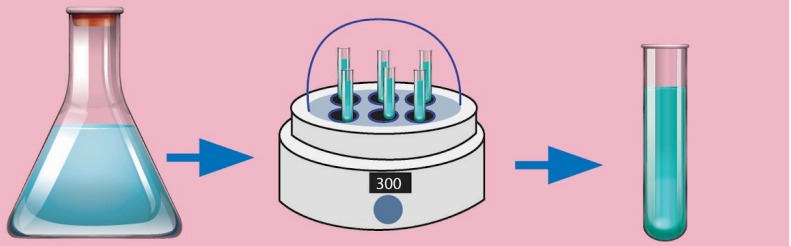
Phage grown with radioactive sulfur (^{35}S) → Centrifuge → Radioactivity in supernatant

Conclusion : Protiens are not genetic material

Experiment 2 : Testing DNA



Phage DNA radiolabelled → Bacteria infected → Radioactivity enters cells



Phage grown with radioactive phosphorus(^{32}P) → Centrifuge → Radioactivity in pellet

Conlusion : DNA is the genetic material

The Structure of DNA

What do you know about the structure of DNA?

DNA Structure

- Once it was determined that DNA was the genetic material, a new set of questions began to emerge.
- One of the first was how cells are able to accurately replicate their DNA to be passed on during cell division (HEREDITARY?)
- Answering this question required an understanding of the *three-dimensional structure of DNA*
- DNA Double Helix Discovery

https://www.hhmi.org/biointeractive/double-helix

Chargaff's Rules Reveal That A = T and G = C

- Erwin Chargaff was interested in the base composition of DNA and used chromatographic methods to separate and quantify the relative amounts of the four bases
- He showed that the DNA from *different cells of a given species has the same percentage of each of the four bases*
- The base composition varies among species

Table 16-1 DNA Base Composition Data That Led to Chargaff's Rules

	Number of Each Type of Nucleotide*				Nucleotide Ratios**		
Source of DNA	**A**	**T**	**G**	**C**	**A/T**	**G/C**	**(A + T)/(G + C)**
Bovine thymus	28.4	28.4	21.1	22.1	1.00	0.95	1.31
Bovine liver	28.1	28.4	22.5	21.0	0.99	1.07	1.30
Bovine kidney	28.3	28.2	22.6	20.9	1.00	1.08	1.30
Bovine brain	28.0	28.1	22.3	21.6	1.00	1.03	1.28
Human liver	30.3	30.3	19.5	19.9	1.00	0.98	1.53
Locust	29.3	29.3	20.5	20.7	1.00	1.00	1.41
Sea urchin	32.8	32.1	17.7	17.3	1.02	1.02	1.85
Wheat germ	27.3	27.1	22.7	22.8	1.01	1.00	1.19
Marine crab	47.3	47.3	2.7	2.7	1.00	1.00	17.50
Aspergillus (mold)	25.0	24.9	25.1	25.0	1.00	1.00	1.00
Saccharomyces cerevisiae (yeast)	31.3	32.9	18.7	17.1	0.95	1.09	1.79
Clostridium (bacterium)	36.9	36.3	14.0	12.8	1.02	1.09	2.73

*The values in these four columns are the average number of each type of nucleotide found per 100 nucleotides in DNA.
**The A/T and G/C ratios are not all exactly 1.00 because of experimental error.

Chargaff's Most Striking Observation

- Chargaff observed that for all DNA samples examined, the number of A = the number of T, and the number of G = the number of C
- These are called **Chargaff's rules**
- The significance was not understood until Watson and Crick proposed the double-helix model for DNA structure!

Linus Pauling's Work on DNA

- Linus Pauling identified the alpha strand and beta sheet structure of proteins. Then he turned his focus onto another cellular polymer, DNA.

A PROPOSED STRUCTURE FOR THE NUCLEIC ACIDS

By Linus Pauling and Robert B. Corey

Gates and Crellin Laboratories of Chemistry,* California Institute of Technology

Communicated December 31, 1952

The nucleic acids, as constituents of living organisms, are comparable in importance to the proteins. There is evidence that they are involved in the processes of cell division and growth, that they participate in the transmission of hereditary characters, and that they are important constituents of viruses. An understanding of the molecular structure of the nucleic acids should be of value in the effort to understand the fundamental phenomena of life.

We have now formulated a promising structure for the nucleic acids, by making use of the general principles of molecular structure and the available information about the nucleic acids themselves. The structure is not a vague one, but is precisely predicted; atomic coordinates for the principal atoms are given in table 1. This is the first precisely described structure for the nucleic acids that has been suggested by any investigator. The structure accounts for some of the features of the x-ray photographs; but detailed intensity calculations have not yet been made, and the structure cannot be considered to have been proved to be correct.

The Formulation of the Structure.—Only recently has reasonably complete information been gathered about the chemical nature of the nucleic acids. The nucleic acids are giant molecules, composed of complex units. Each unit consists of a phosphate ion, HPO_4^{--}, a sugar (ribose in the ribonucleic

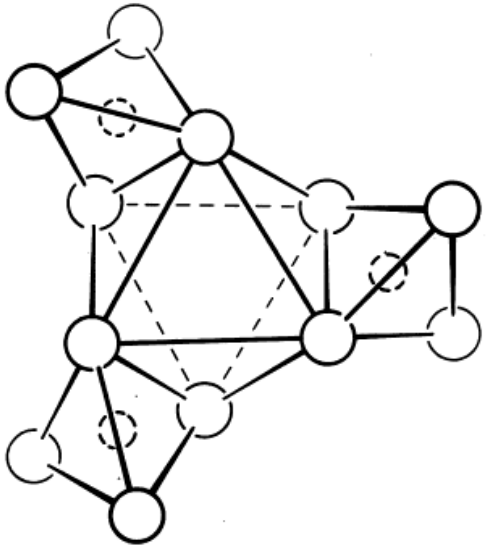


FIGURE 1

A group of three phosphate tetrahedra near the axis of the nucleic acid molecule. Oxygen atoms are indicated by full circles and phosphorus atoms by dashed circles.

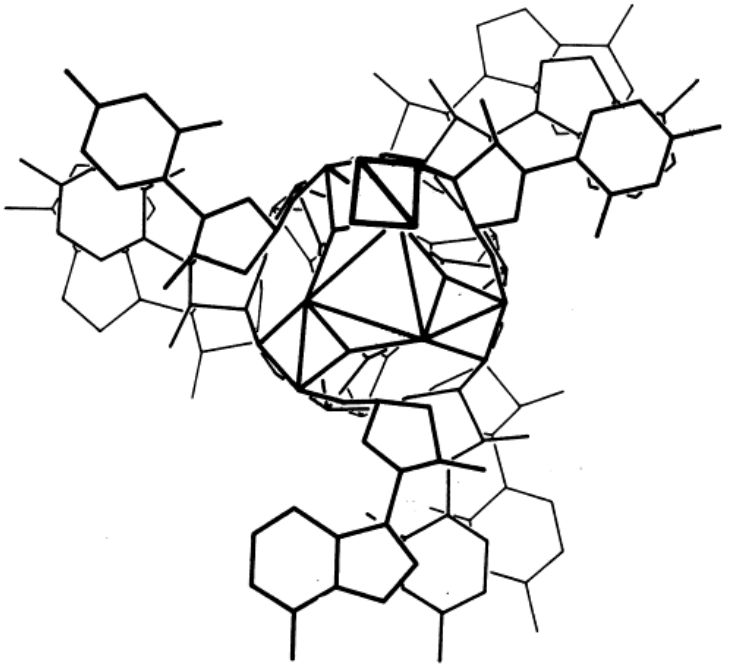


FIGURE 6

Plan of the nucleic acid structure, showing several nucleotide residues.

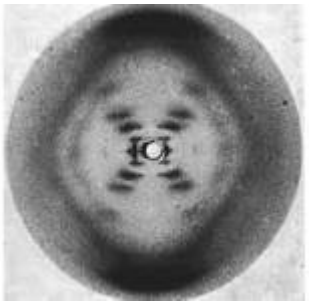
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1063734/?page=1

Watson and Crick Discovered That DNA Is a Double Helix

- James Watson and Frances Crick built wire models to try to determine the structure of DNA that agreed with everything known about DNA
- It was known that DNA had a sugar phosphate backbone with nitrogenous bases attached to each sugar
- It was known that at physiological pH, the bases would be able to form hydrogen bonds with each other

The Double Helix Model

- The critical evidence came from X-ray diffraction data produced by Rosalind Franklin
- It revealed that DNA was a long thin helical molecule

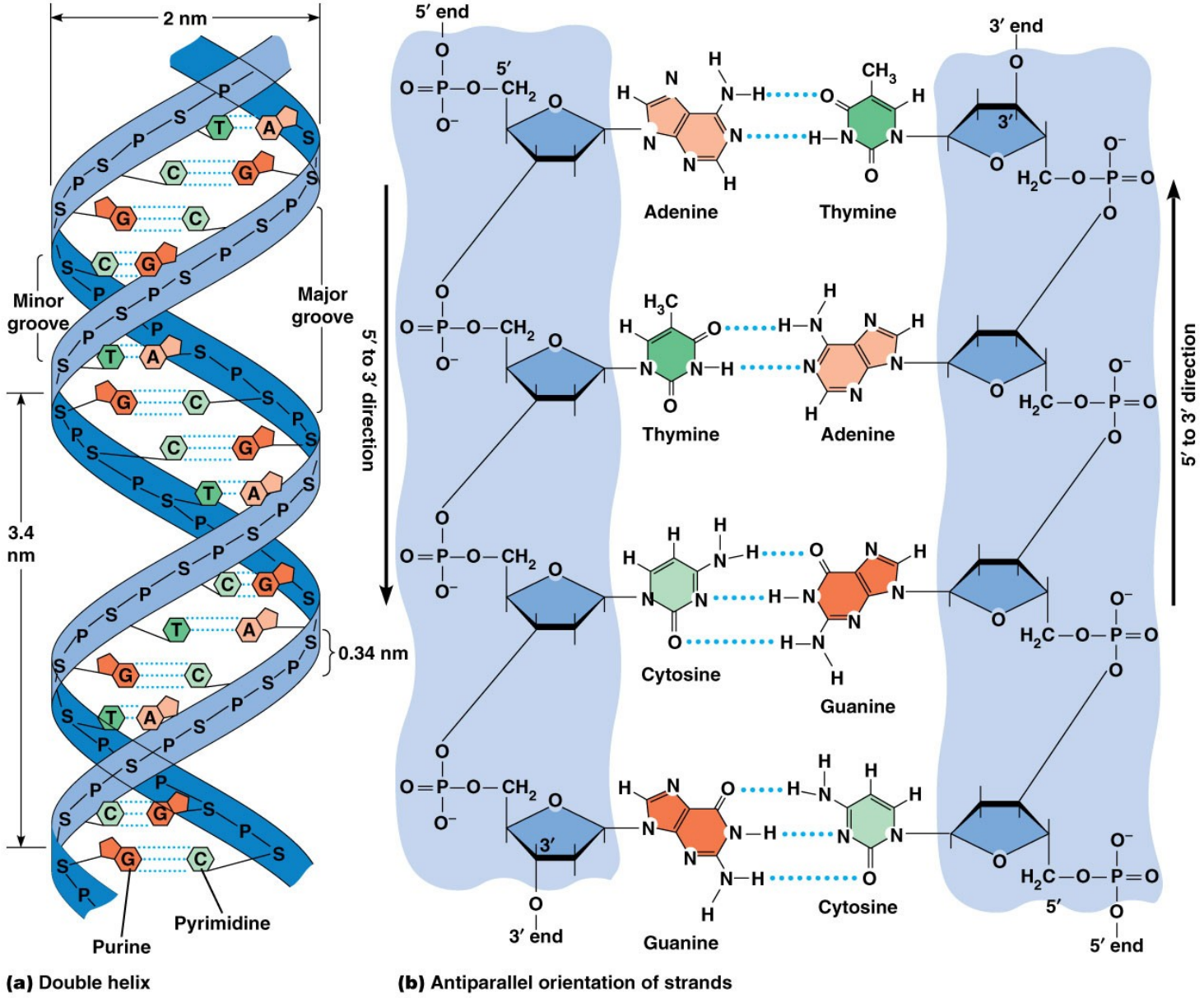


"Photo 51"

- Based on this information and other observations, Watson and Crick produced the **double helix** model

The Double Helix Model

- The purine-pyrimidine pairing is consistent with Chargaff's rules
- The two strands are held together by hydrogen bonding between bases on opposite strands
- The hydrogen bonds fit within the helix only when they form between **complementary** bases: adenine with thymine and guanine with cytosine


2 nm
Minor groove
Major groove
3.4 nm
0.34 nm
Purine
Pyrimidine
(a) Double helix
5′ end
3′ end
5′ to 3′ direction
Adenine
Thymine
Thymine
Adenine
Cytosine
Guanine
Guanine
Cytosine
5′ to 3′ direction
(b) Antiparallel orientation of strands

Key Features of DNA Structure

- The strands of DNA are twisted around each other so that there is a *major groove* and a *minor groove*
- The phosphodiester (covalent) bonds that join the 5′ carbon of one nucleotide to the 3′ carbon of the next are oriented in *opposite* directions in the two DNA strands
- This is called *antiparallel* orientation

Measuring DNA Length

- DNA length is measured in **base pairs** (**bp**)
- Larger stretches are measured in multiples of a single base pair—for example, the **kilobase** (**kb**) is 1000 bp

Replication of Genetic Information

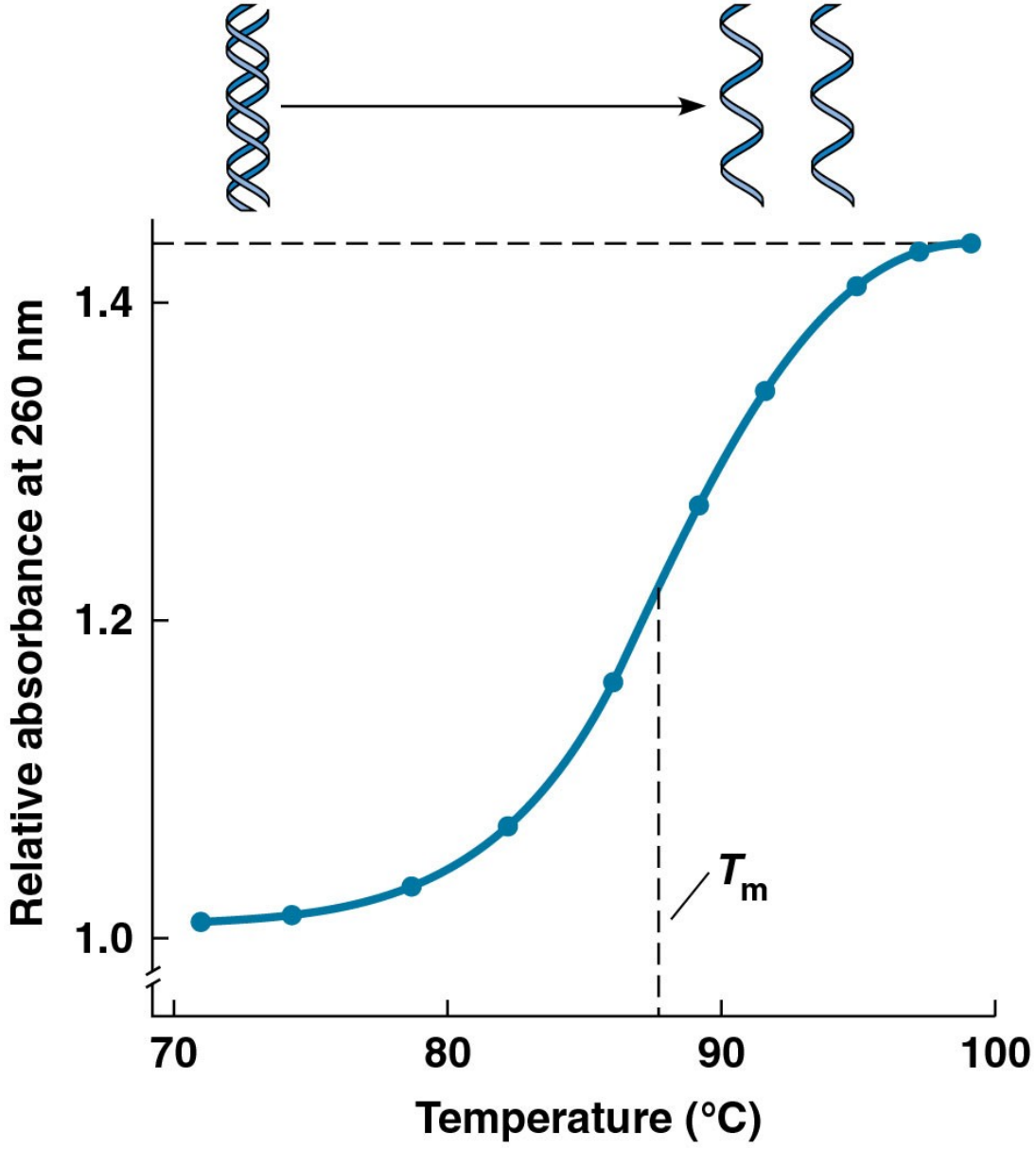
- The most important aspect of the double helix model was that it suggested **a mechanism for replication of DNA**
- The two strands could separate so that each could act as a *template* to dictate synthesis of a new complementary strand

The Two Strands of a DNA Double Helix Can Be **Denatured and Renatured**

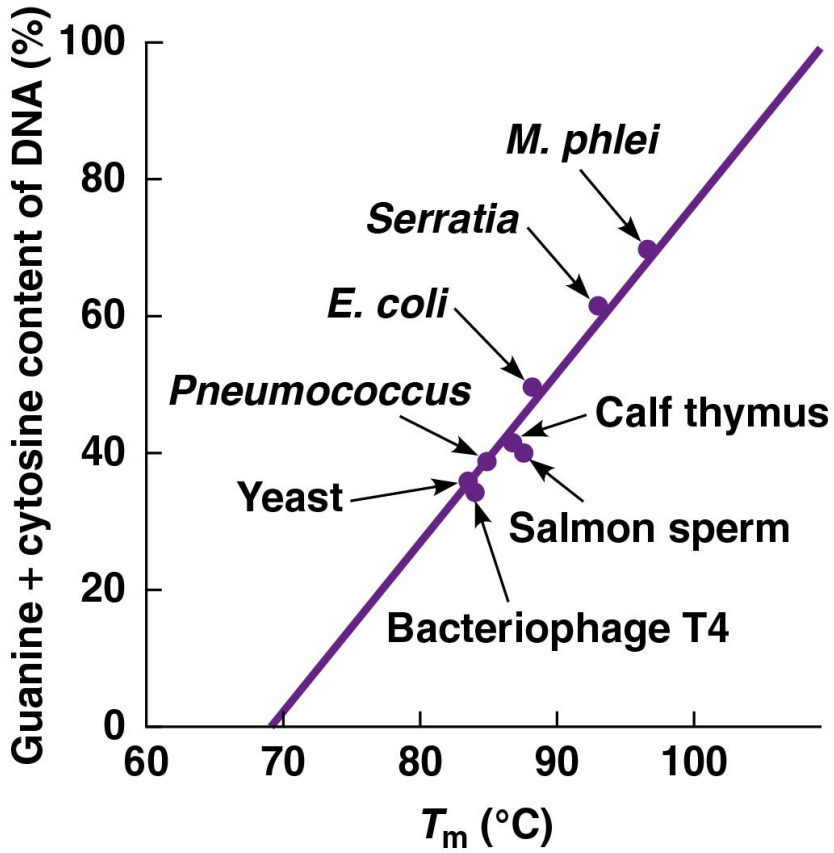
- DNA strands are bound together by relatively weak noncovalent bonds
- Strand separation (**DNA denaturation** or **melting**) can be induced experimentally by raising temperature or pH
- The process can be monitored because single- and double-stranded DNA differ in light absorption

Denaturation

- All DNA absorbs light, with a maximum around
  260 nm
- As the strands separate, the absorbance increases rapidly
- The temperature at which half of the absorbance change is reached is called the **DNA melting temperature, T_m**; the T_m varies depending on A-T content

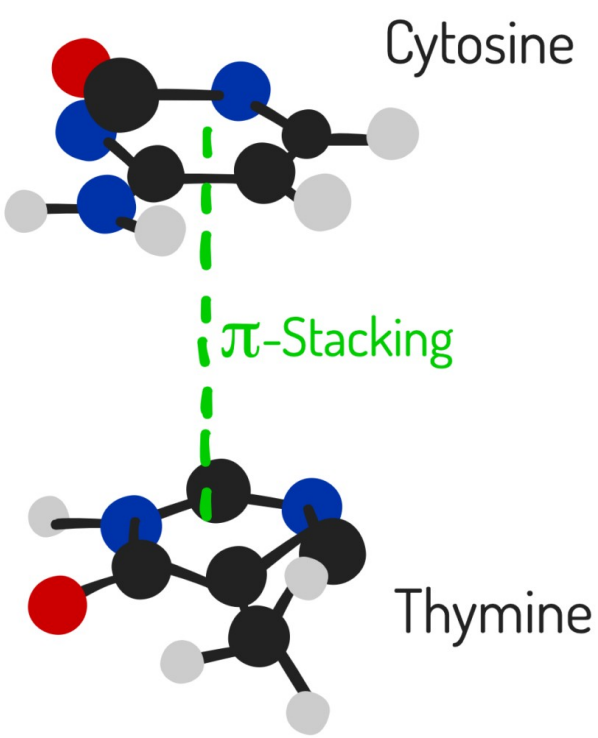

Relative absorbance at 260 nm
1.4
1.2
1.0
70
80
90
100
Temperature (°C)
T_m

Increased G-C content of DNA results in higher Tm



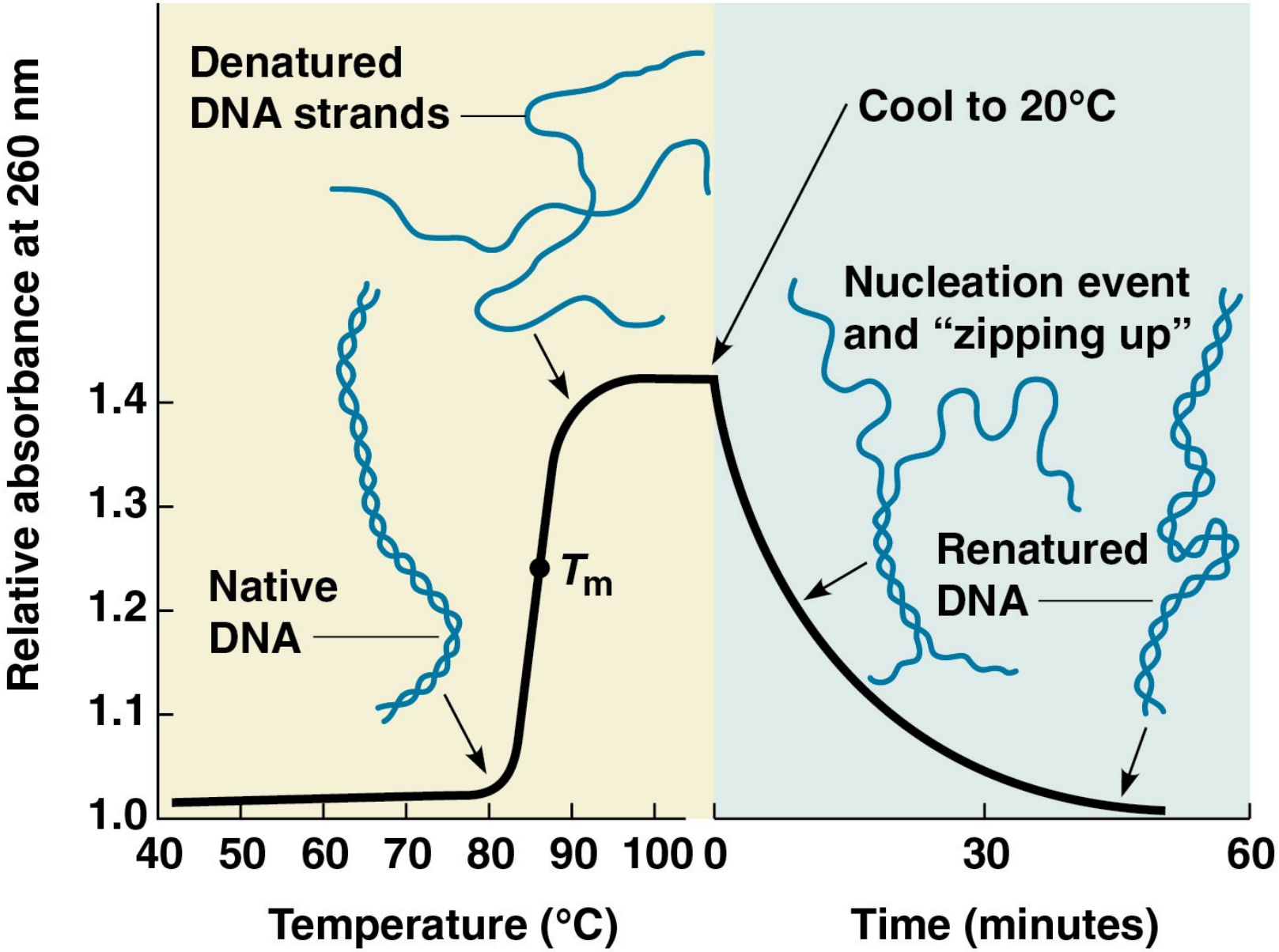
Base Stacking

- A major interaction within each single strand stabilizes the double helix
- **Base stacking** involves interactions between adjacent aromatic rings by hydrophobic and van der Waals interactions



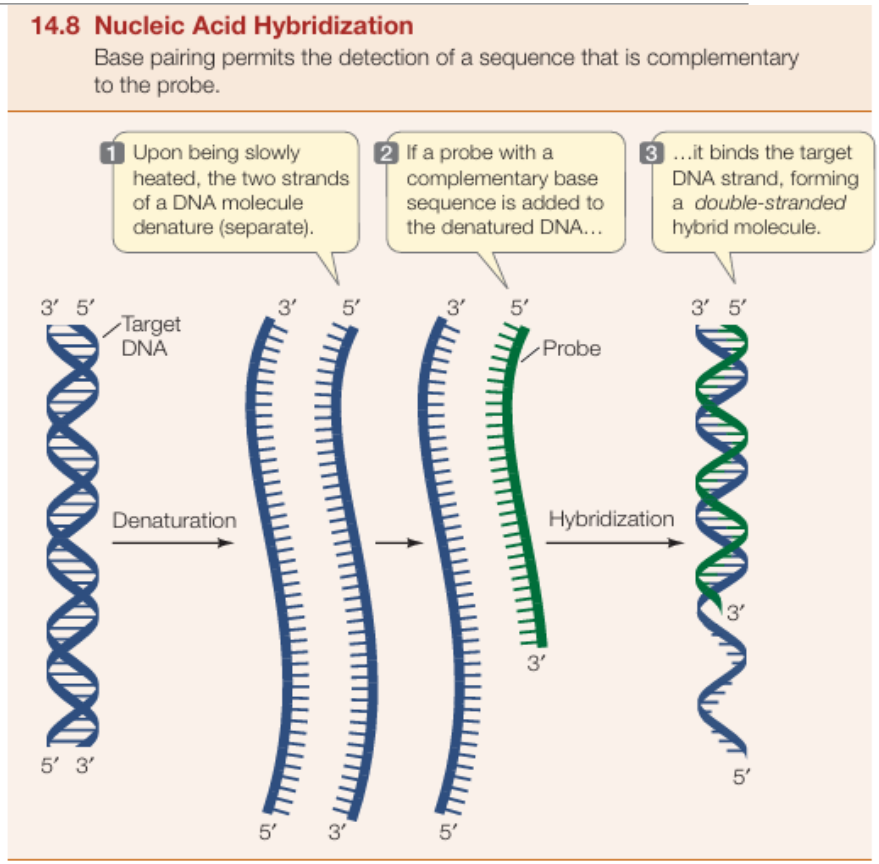
Renaturation

- Reformation of the DNA double helix is called **DNA renaturation** (**reannealing**); it is accomplished by lowering the temperature to permit hydrogen bonds to reform
- This process dependent on time and the DNA concentration.
- If the DNA is very long and complex, renaturation may not be precise. Thus, the rate of renaturation can give idea about the complexity of a DNA molecule.


Relative absorbance at 260 nm
1.4
1.3
1.2
1.1
1.0
Denatured
DNA strands
Native
DNA
T_m
Cool to 20°C
Nucleation event
and "zipping up"
Renatured
DNA
40 50 60 70 80 90 100
Temperature (°C)
0 30 60
Time (minutes)

Renaturation

- In **nucleic acid hybridization,** nucleic acids can be identified based on sequence
- Denatured DNA is incubated with a purified single-stranded DNA (a **probe**) with a sequence complementary to the sequence one is trying to detect

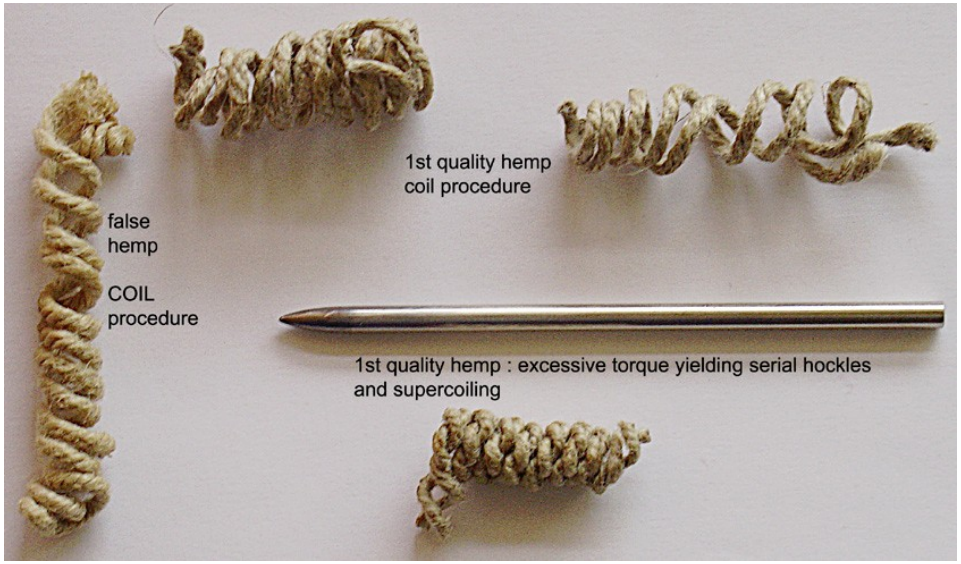


Packaging the DNA

DNA Packaging

- PROBLEM: Very long molecules of DNA must be fit into the cell and, in the case of eukaryotes, into the nucleus
- 2 meters of DNA into 6 micron cell nucleus. In comparison, average human hair is 50 micron in diameter.
- DNA packaging is a challenge for all forms of life

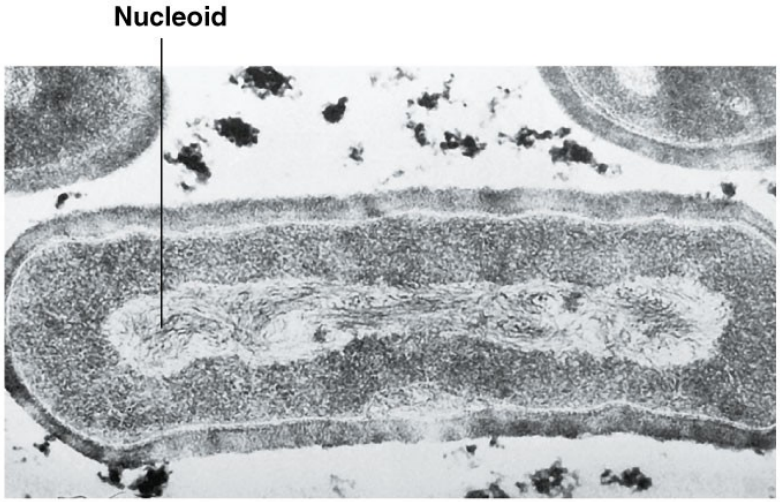
How do you package DNA into nucleus?





Bacteria Package DNA in Bacterial Chromosomes and Plasmids

- The main bacterial genome is called the **bacterial chromosome**
- Bacteria have single, multiple, linear, or circular chromosomes depending on the species, but a single circular chromosome is most common
- The loops of bacterial DNA are held in place by RNAs and proteins



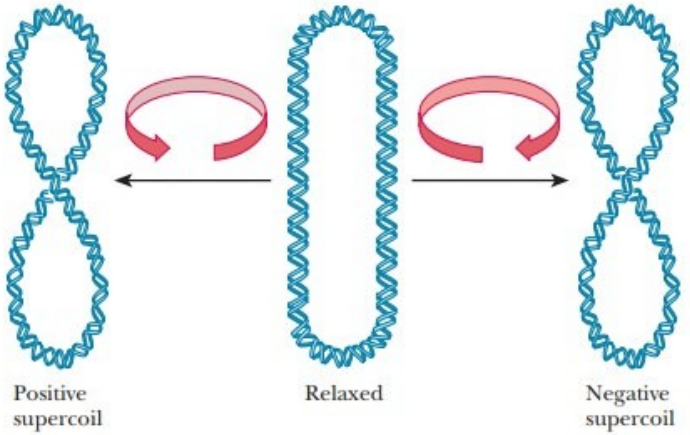
(a) Bacterial nucleoid



(b) Released DNA

Supercoiling

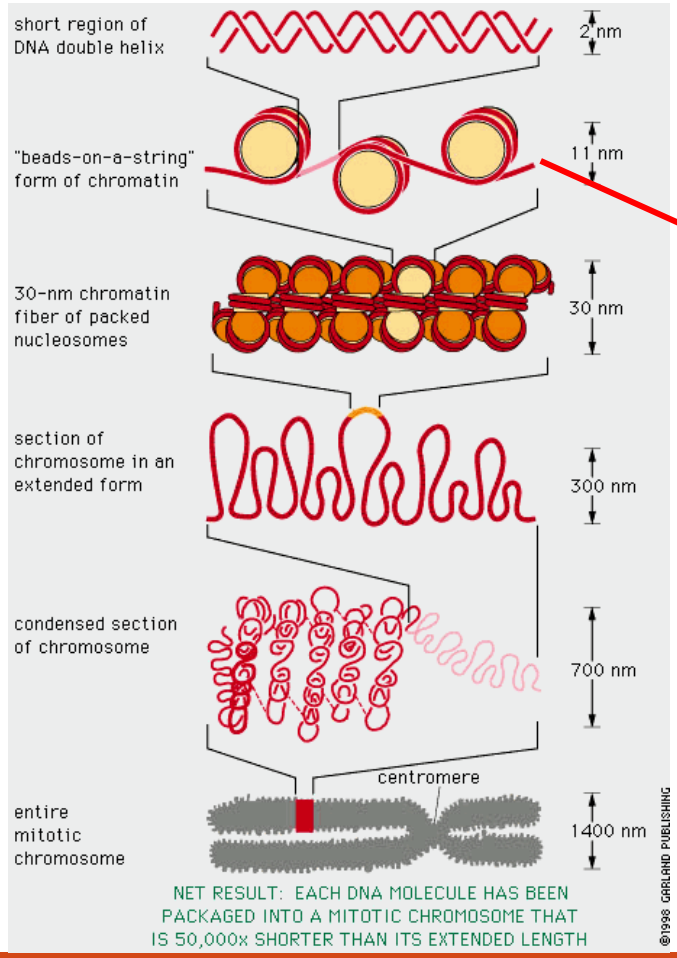
- Supercoiling occurs in both linear and circular DNA molecules but is more easily studied in circular DNA
- A DNA molecule can go back and forth between the supercoiled state and the nonsupercoiled, or *relaxed*
- Extensive supercoiling helps make chromosomal DNA more compact

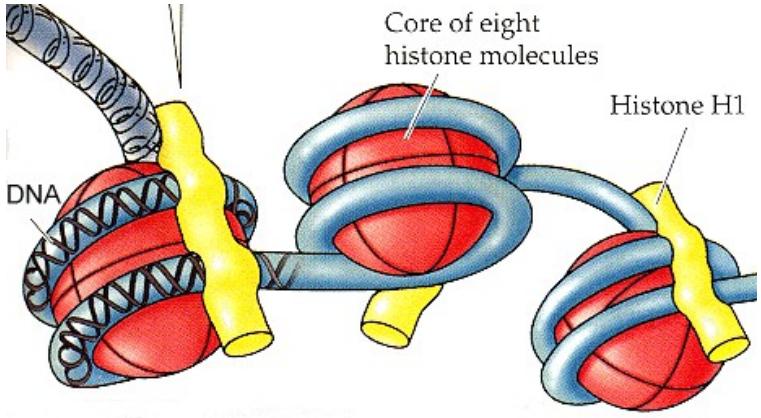


Eukaryotes Package DNA in Chromatin and Chromosomes

- In eukaryotes, there is more DNA per cell, and it interacts with specific proteins called histones
- When bound to proteins, DNA is called **chromatin**
- At the time of division, the chromatin fibers condense into a more compact structure, the **chromosome**

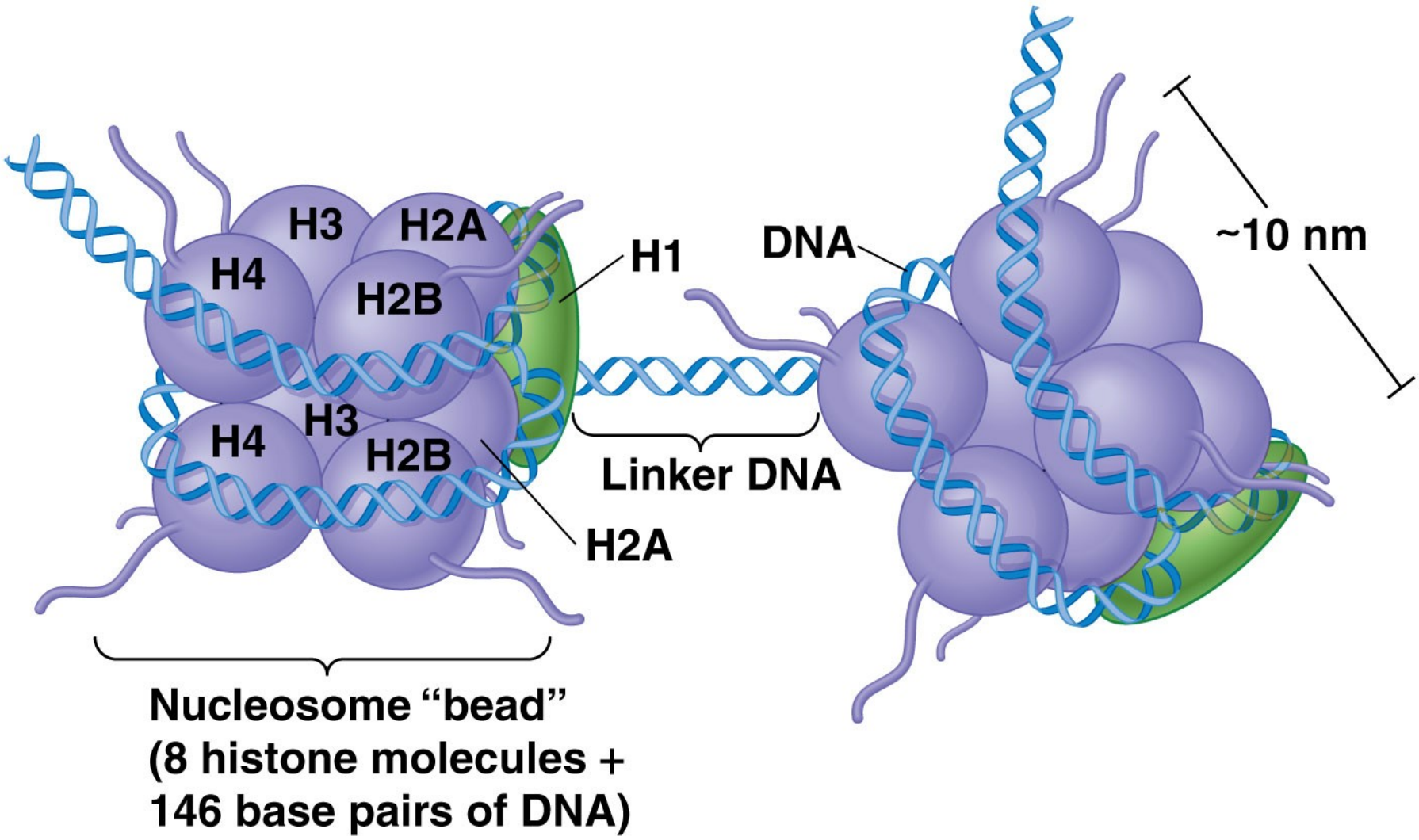
Levels of compaction in eukaryotic chromosome





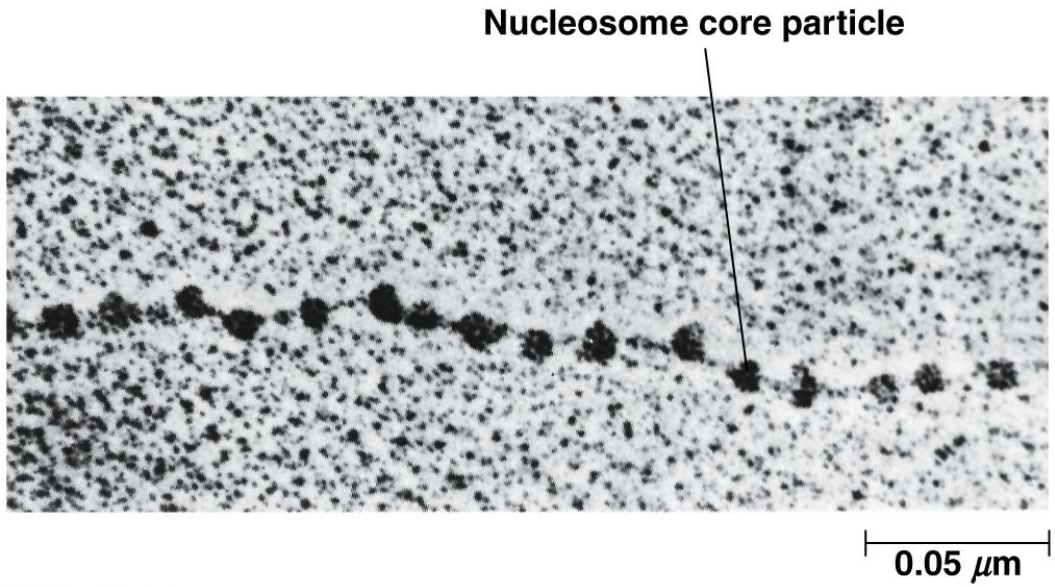
Histones

- **Histones** are a group of small basic proteins with high lysine and arginine content
- The negatively charged DNA binds stably to the positively charged proteins
- The mass of histones in a chromosome is approximately equal to the mass of the DNA


H3
H2A
H4
H2B
H1
DNA
~10 nm
H4
H3
H2B
Linker DNA
H2A
Nucleosome "bead"
(8 histone molecules +
146 base pairs of DNA)

Types of Histones

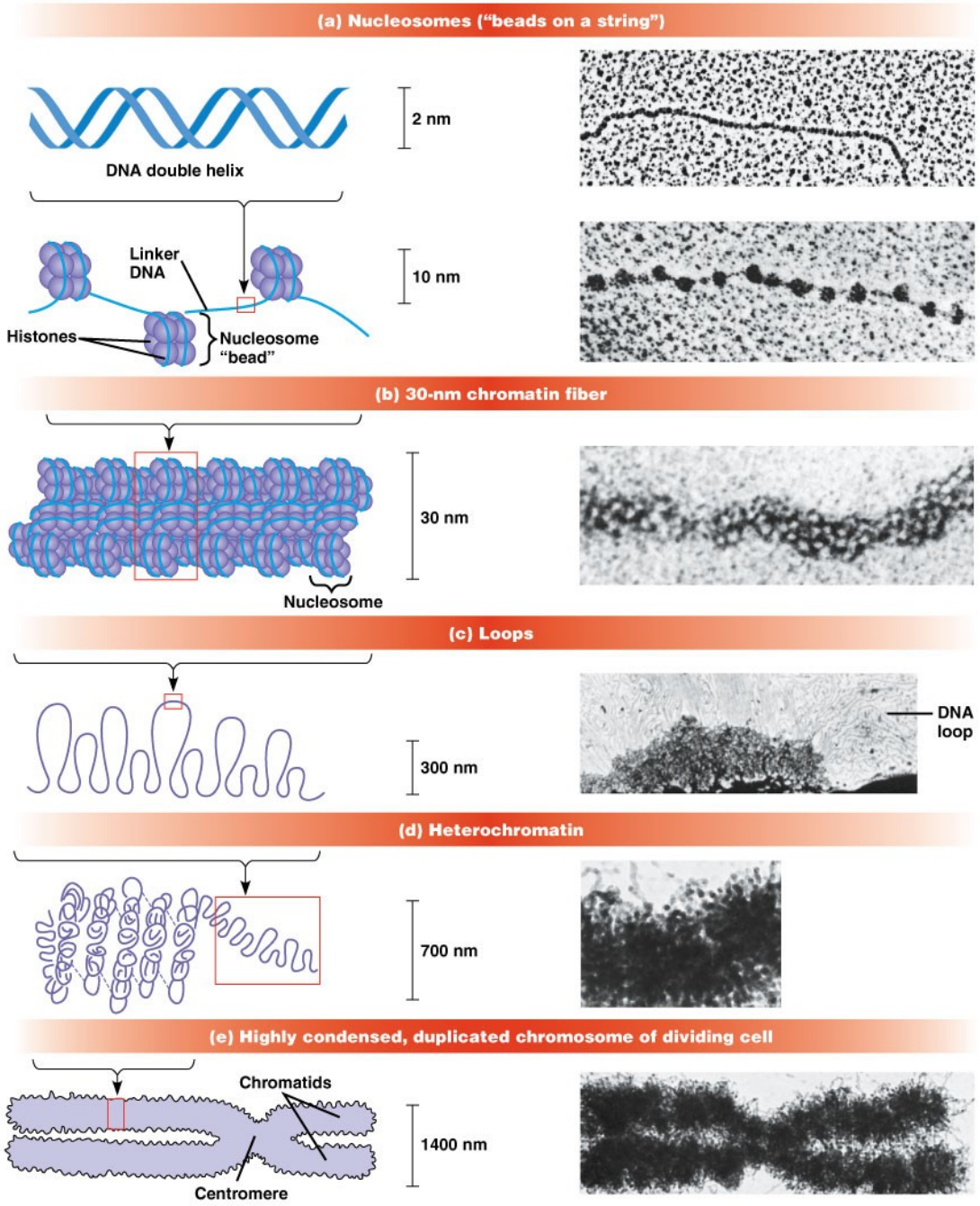
- Two each of H2A, H2B, H3, and H4 (total of 8) form ‘nucleosome’ structure
- Histones are conserved in all eukaryotes



Chromatin Remodeling Proteins Can Alter Chromatin Packing

- Each histone has a protruding tail that can be tagged by the addition of methyl, acetyl, phosphate, or other groups
- Various combinations of these tags create a **histone code**
- http://www.rcsb.org/pdb/explore/jmol.do?structureId=1eqz&bionumber=1&jmolMode=HTML5

Diagram and TEM images showing the progressive stages of DNA coiling and folding

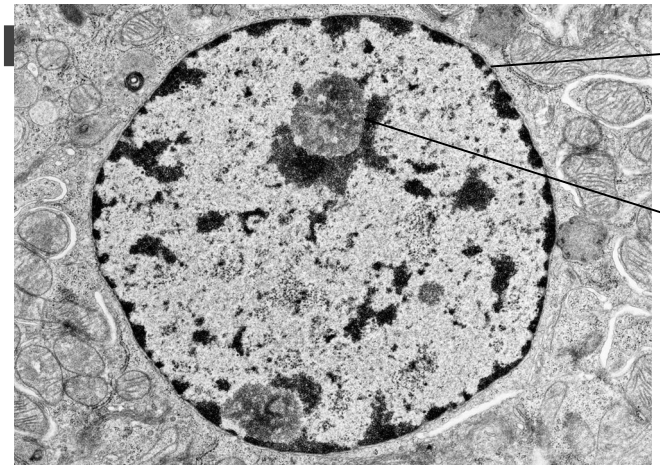


Transcription and Packaging

- Transcriptionally active DNA is less tightly packed than inactive DNA
- This makes sense because a “looser” packaging would allow easy access by proteins involved in gene transcription
- Gene transcription is also influenced by chemical modifications of histone proteins.

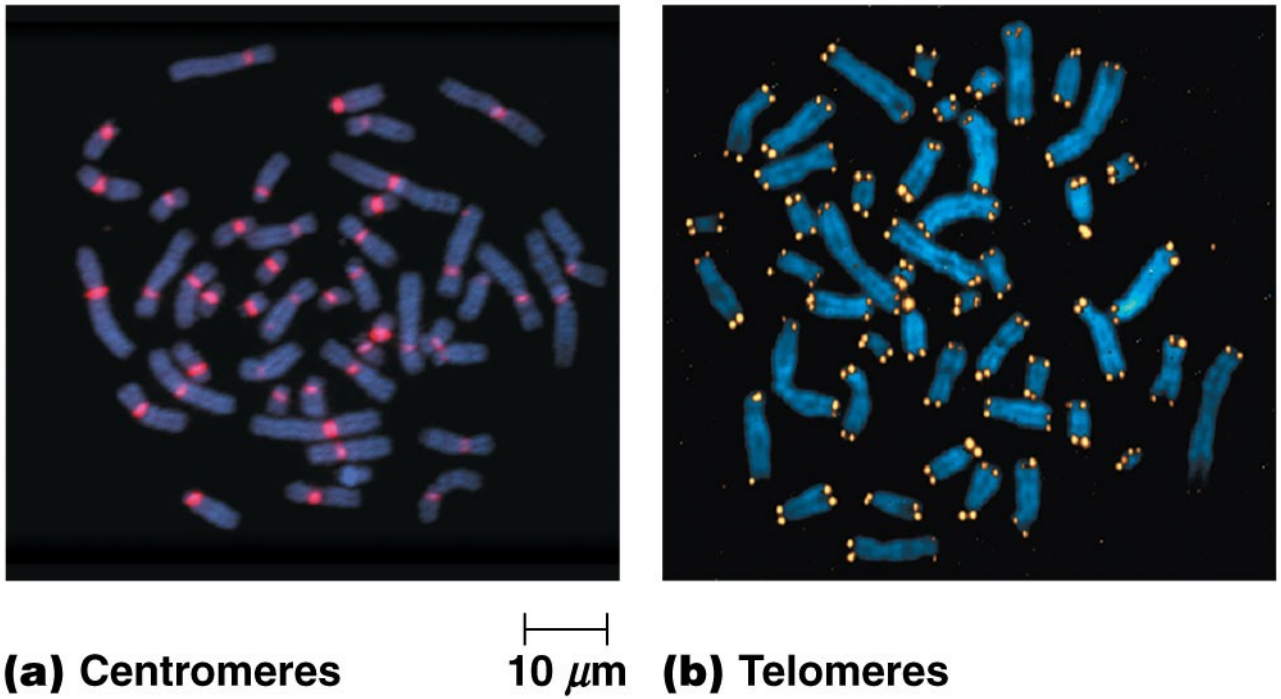
Chromosomal DNA Contains Euchromatin and Heterochromatin

- Sections of chromatin so highly compacted that they show up as dark spots in micrographs are called **heterochromatin**
- More loosely packed, diffuse chromatin is called **euchr**

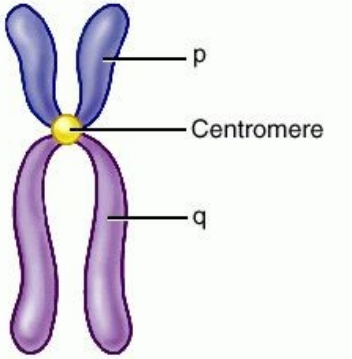


Some Heterochromatin Plays a Structural Role in Chromosomes

Two important types of heterochromatin are **centromeres** and **telomeres**



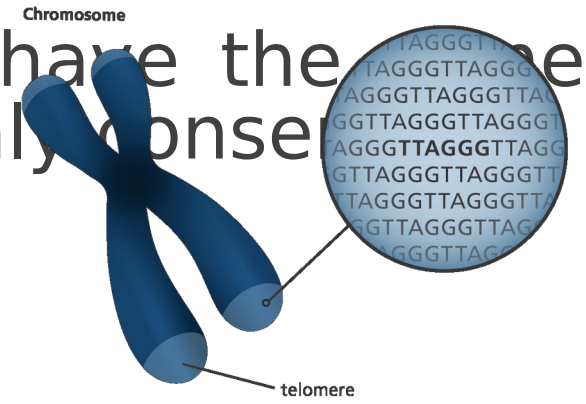
Centromeres



- Centromeres are characterized by highly repetitive DNA sequences (CEN sequences)
- Centromere DNA is bound by a complex of proteins and serves important functions
- Centromeres maintain sister chromatid cohesion during mitosis and meiosis
- They also serve as sites of kinetochores, crucial for attaching

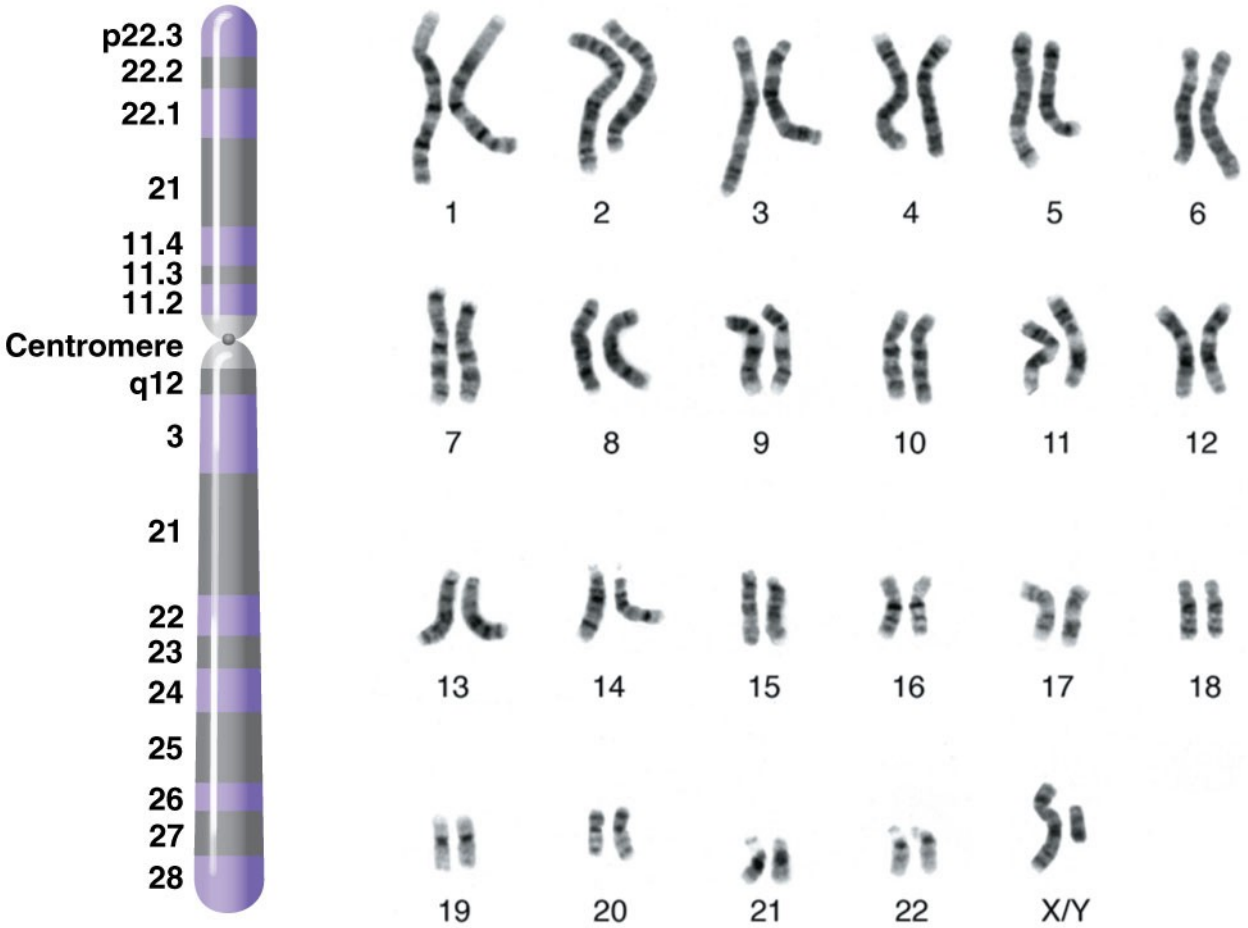
Telomeres

- Telomeres are found at the tips of chromosomes
- They contain highly repetitive DNA sequences
- Telomeres protect chromosome ends from degradation during each round of DNA replication
- All vertebrates studied so far have the same repeat sequence (TTAGGG), highly conserved
- 250-1500 copies



Chromosomes Can Be Identified by Unique Banding Patterns

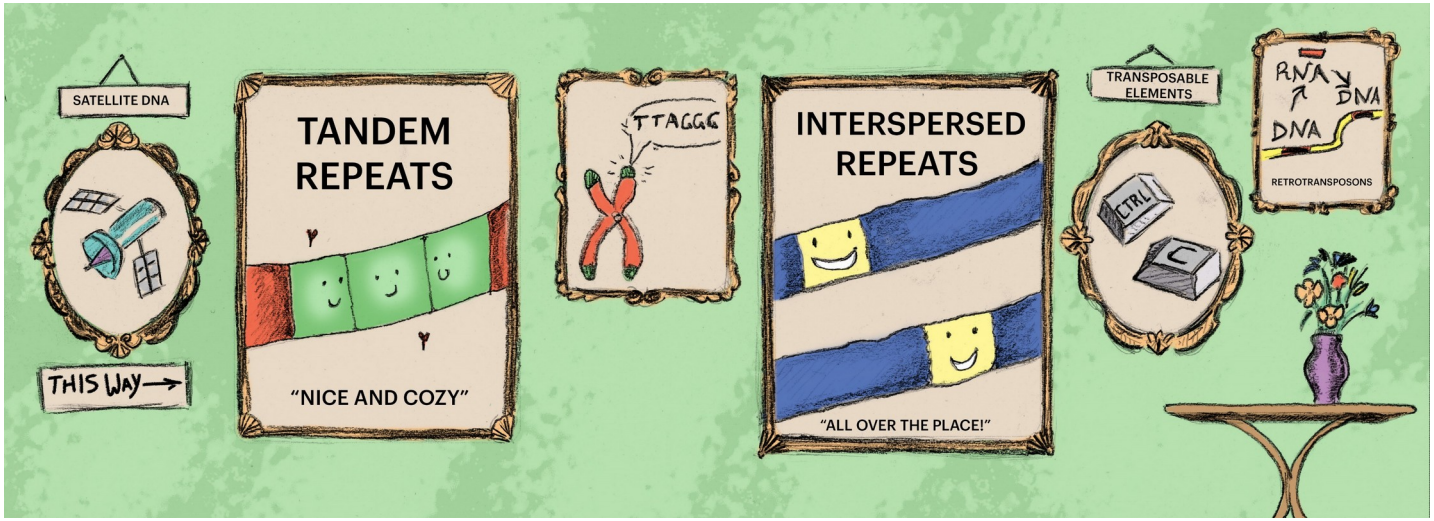
- Mitotic chromosomes viewed under the light microscope can be distinguished by size and position of the centromere
- Similar-sized chromosomes can be distinguished by their banding patterns in response to stains
- A common stain is Giemsa, which causes light- and dark-staining chromosome bands called *G bands*



(a) G-bands on the human X chromosome

(b) Giemsa-stained chromosomes from a human male

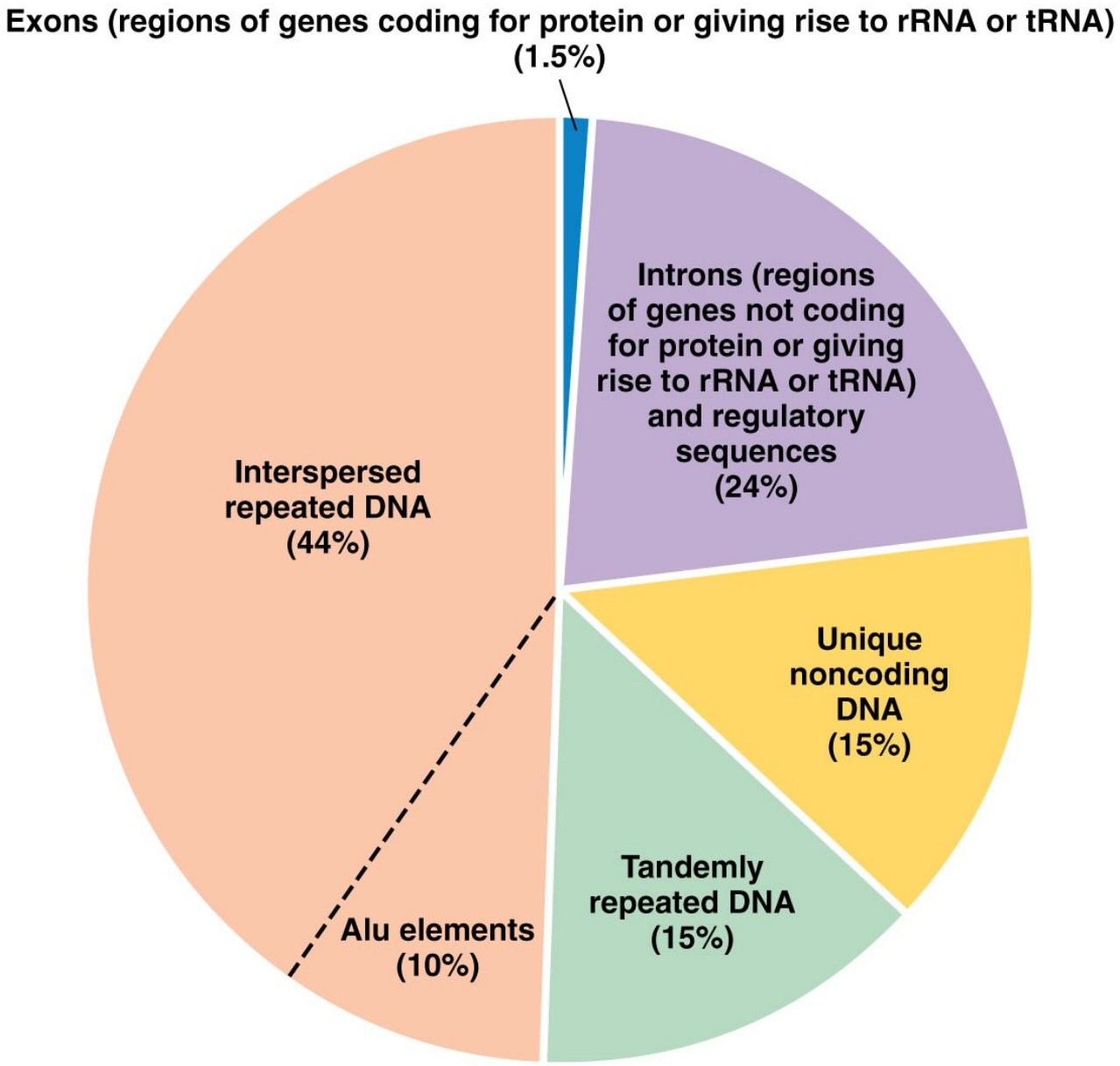
Genomic DNA has many repeat sequences



Eukaryotic DNA contains many repeat sequences

Table 16-2 Categories of Repeated Sequences in Eukaryotic DNA

I. Tandemly repeated DNA, including satellite DNA	
10–15% of most mammalian genomes is this type of DNA.	
Length of each repeated unit:	1–2000 bp; typically 5–10 bp for highly repetitive DNA
Number of repetitions per genome:	10^2–10^5
Arrangement of repeated units:	Tandem
Total length of satellite DNA at each site:	
Regular satellite DNA:	10^5–10^7 bp
Variable number tandem repeats (VNTRs):	10^2–0^5 bp
Short tandem repeats (STRs):	10^1–10^2 bp
II. Interspersed repeated DNA	
25–50% of most mammalian genomes is this type of DNA.	
Length of each repeated unit:	10^2–10^4 bp
Arrangement of repeated units:	Scattered throughout the genome
Number of repetitions per genome:	10^1–10^6; "copies" not identical


Exons (regions of genes coding for protein or giving rise to rRNA or tRNA)
(1.5%)
Introns (regions of genes not coding for protein or giving rise to rRNA or tRNA) and regulatory sequences
(24%)
Unique noncoding DNA
(15%)
Tandemly repeated DNA
(15%)
Alu elements
(10%)
Interspersed repeated DNA
(44%)